12 **(D)**

Gravitational potential at Y $= \frac{GM}{2L} = \frac{1}{2} \times \frac{GM}{L} = \frac{1}{2} \times (-8) = -4\ kJ\ kg^{-1}$

Change in gravitational potential from X to Y $= (-4) - (-8) = 4\ kJ\ kg^{-1}$

Change in gravitational potential energy of 2 kg moved from X to Y $= 2 \times 4 = 8\ kJ$

13 **(B)**